

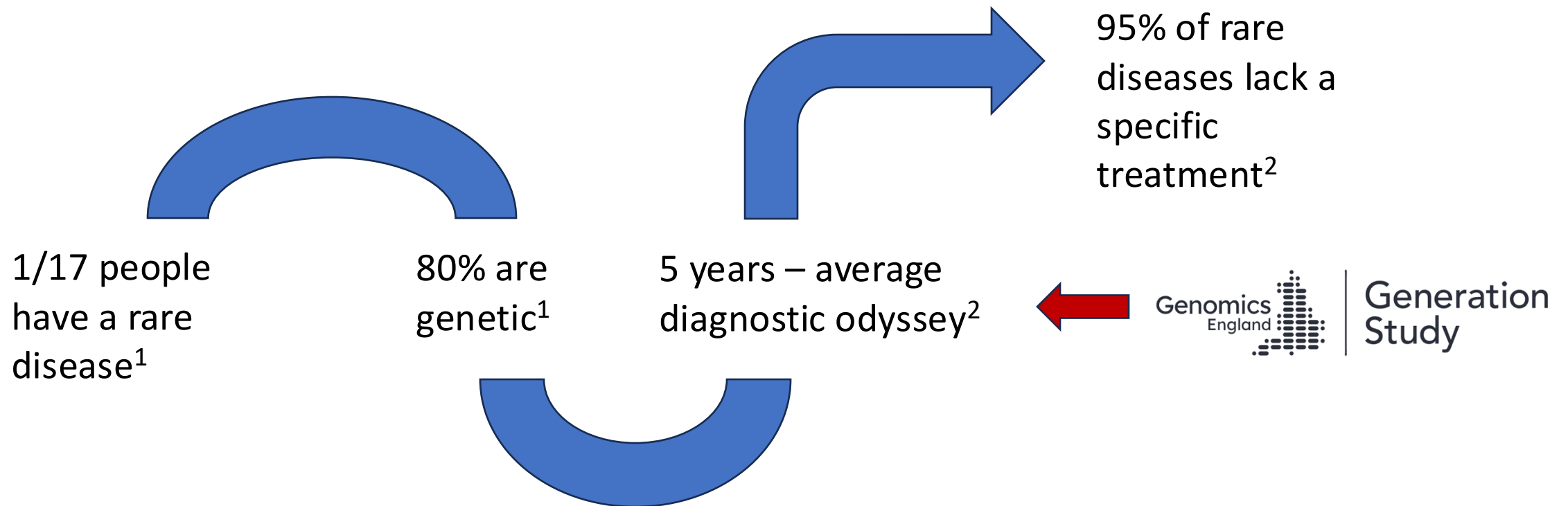
Why MAVEs?

Lizzie Radford

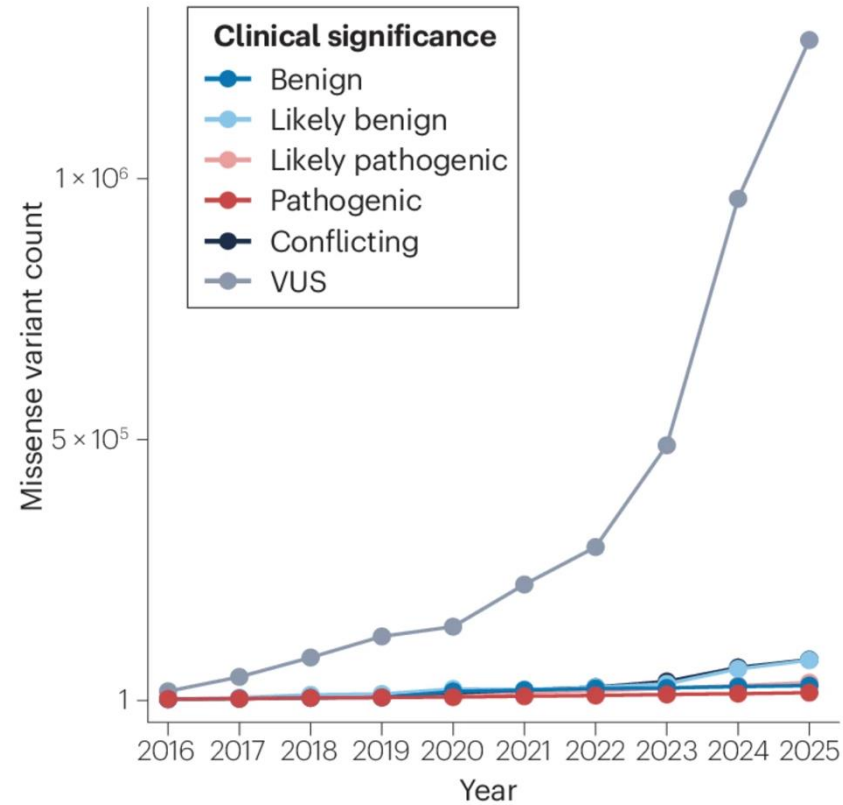
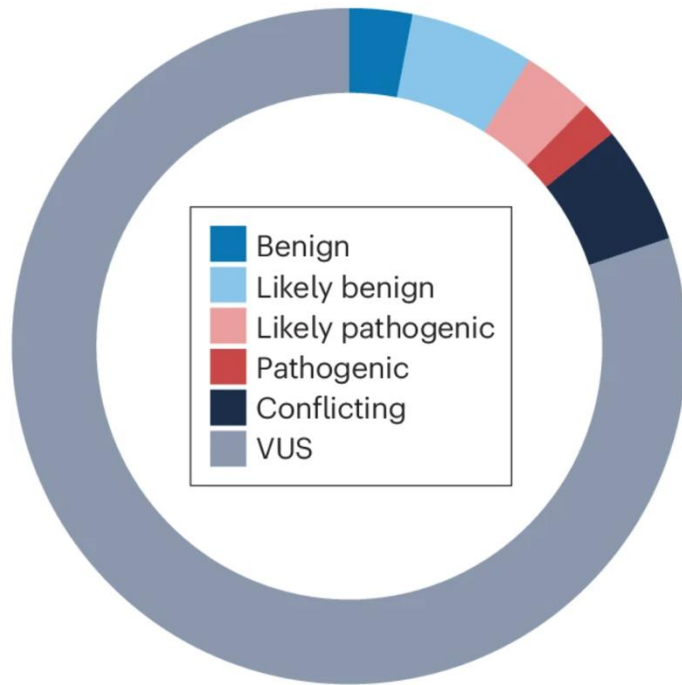
Cambridge University
Hospitals NHS FT

University of Cambridge
Wellcome Sanger Institute

Rare disease – WGS is a standard diagnostic tool



Clinical need: Variants of uncertain significance (VUS) are a major diagnostic barrier



Two key questions in clinic:

1. Is this variant a pathogenic variant?

→ Variant classification (P/LP/VUS/LB/B) – ACMG/ACGS

2. Clinical interpretation / diagnosis - is this variant causing disease/symptoms in my patient? Phenotype is key!

→ NB variable penetrance/expressivity

→ This also affects selection of truth sets for validation of assays

Variant interpretation is hard

Variant seen in healthy populations?

~ 95% of all non-synonymous variants in DD-associated genes have never been observed.

Phenotypic fit
Variant observed in similar patients?

DD phenotypes are indistinct and variable

Variant occurred *de novo*?

Limited weight for non-specific phenotypes

In silico variant effect prediction

Poor specificity

Functional studies

Variants of uncertain significance
Laborious, expensive, slow

How can we do functional assays at scale?



GUIDELINE

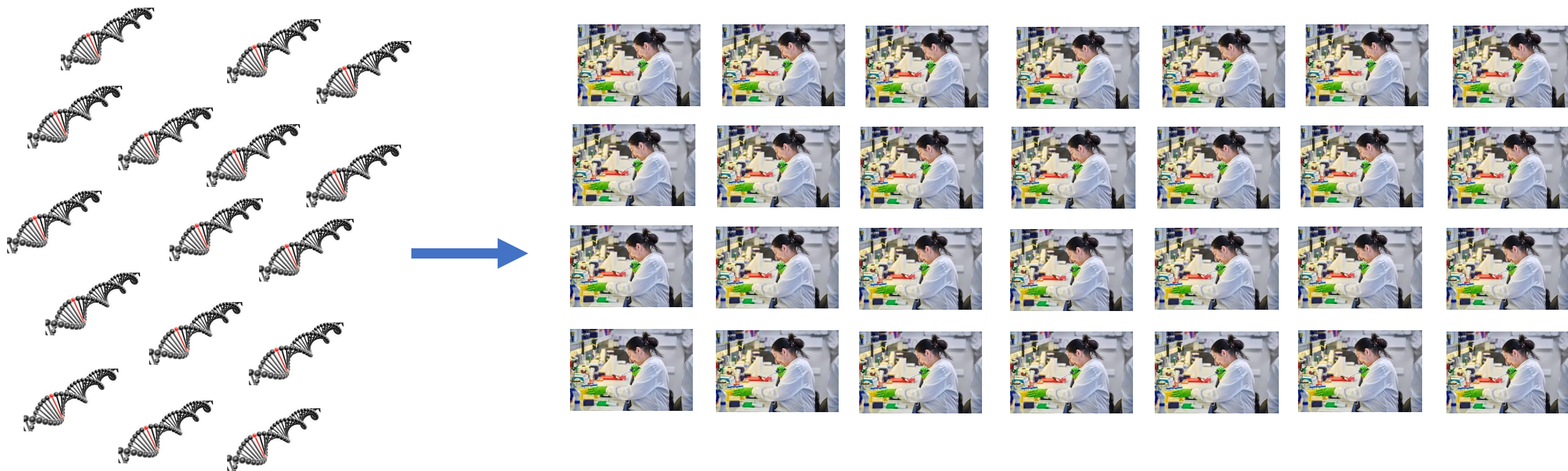
Open Access

Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework

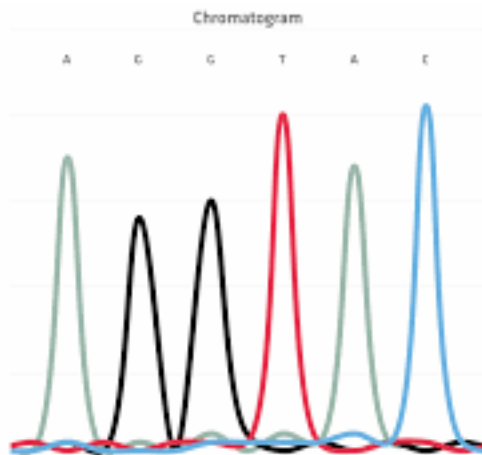


Sarah E. Brnich¹ , Ahmad N. Abou Tayoun², Fergus J. Couch³, Garry R. Cutting⁴, Marc S. Greenblatt⁵, Christopher D. Heinen⁶, Dona M. Kanavy¹, Xi Luo⁷, Shannon M. McNulty¹, Lea M. Starita^{8,9}, Sean V. Tavtigian¹⁰, Matt W. Wright¹¹, Steven M. Harrison¹², Leslie G. Biesecker¹³, Jonathan S. Berg^{1*} and On behalf of the Clinical Genome Resource Sequence Variant Interpretation Working Group

How can we do functional assays at scale?



How can we do functional assays at scale?

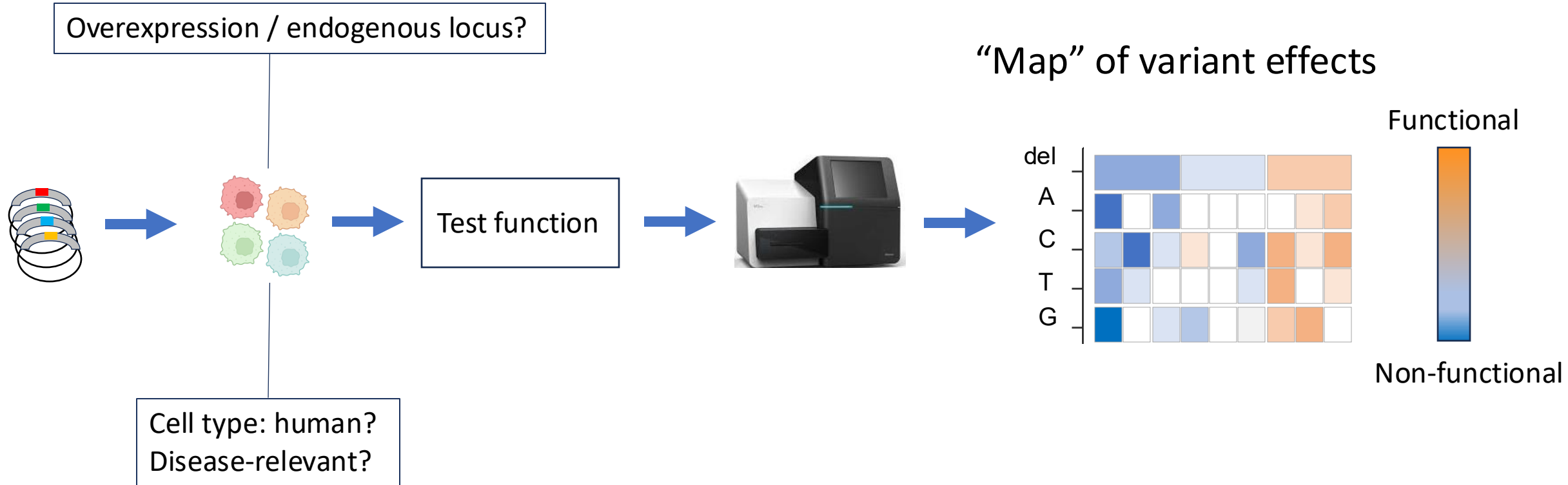


```
ATC CCTCTCCCGCACCTTCGGATA
ATCTT TCCCGCACCTTCGGATA
ATCTTCAACACCGCGC CCGCACCTTCGGATA
ATCTTCAACACCGCGCTCT GCACCTTCGGATA
ATCTTCAACACCGCGCTCTCCCGCACCTTCGGATA
ATCTTCAACACCGCGCTCTCCCGCACCTTCGGATA
ATCTTCAACACCGCGCTCTCCCGCACCTTCGGATA
ATCTTCAACACCGCGCTCTCCCGCACCTTCGGATA
ATCTTCAA
ATCTTCAACACC
ATCTTCAACACCGCGCTCT
```


Multiplexed assays of variant effect

1. What are they?
2. What do I need to think about when designing an assay / when using assay data clinically?
 - a. Disease mechanism
 - b. What variants are testable with the assay
 - c. How to assess evidence for benignity/pathogenicity (intro)
3. What to do when there are multiple MAVEs for your gene?

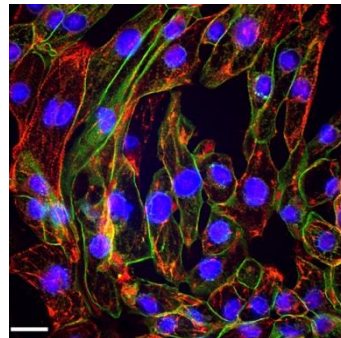
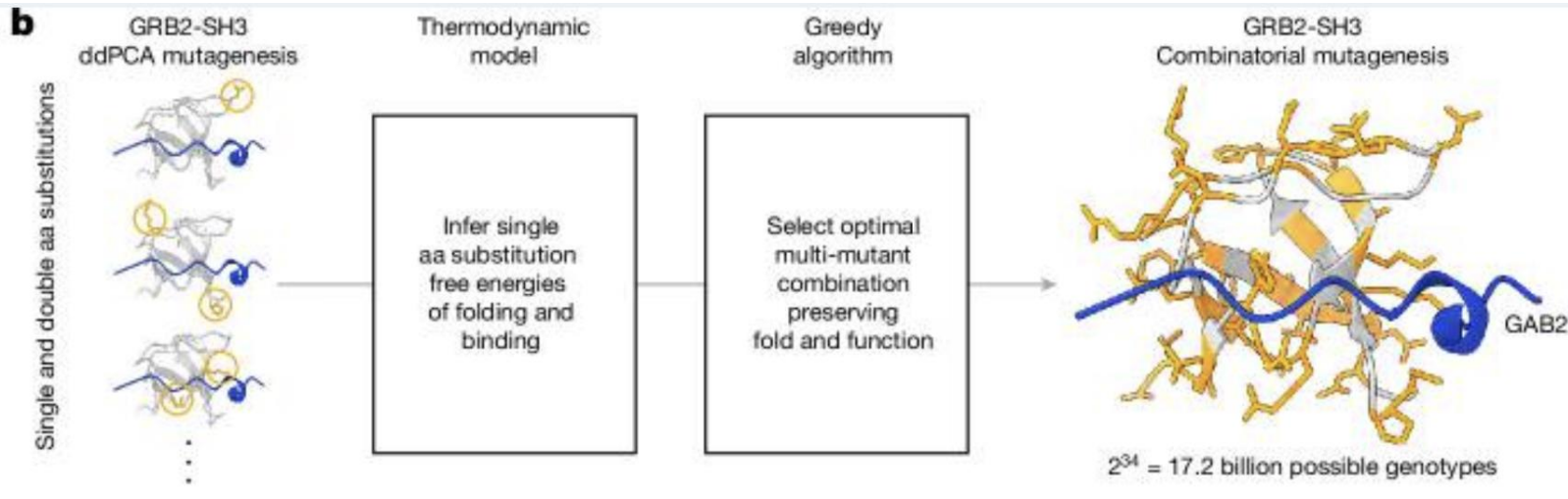
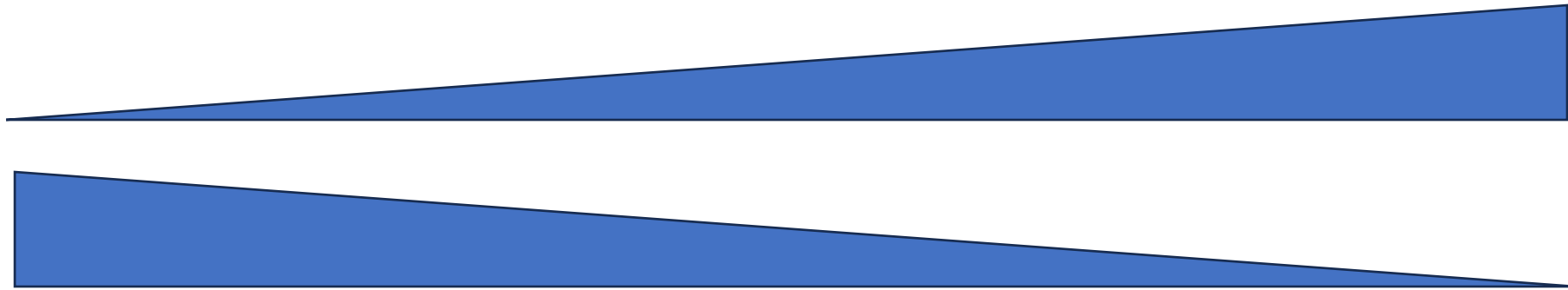
MAVEs – what are they?



MAVEs

Assay
throughput

Disease
relevant
cell type



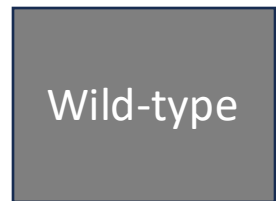
Starita and Findlay labs

Types of MAVEs	Variants tested	Function tested
----------------	-----------------	-----------------

- | | | |
|--|--------------------------|------------------------|
| 1. Edit endogenous locus | All <i>can</i> be tested | Depends on readout |
| 2. Reporter assays | Construct dependent | Abundance/localisation |
| 3. (Over)express a cDNA | Missense | Depends on readout |
| 4. (Over)express a protein domain / fragment | Missense | Abundance/stability |

Matching MAVE to disease mechanism is crucial

Is this functional assay an appropriate measure of variant effect for the condition that I am considering/want to generate data for?

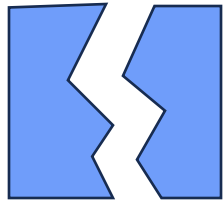


Loss of function



Dominant negative

Assess the utility of MAVE evidence for benignity and pathogenicity separately

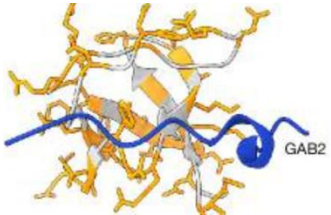


LOF



DN

MAVE measures LOF, condition is caused by both LOF/ DN variants



$2^{34} = 17.2$ billion possible genotypes

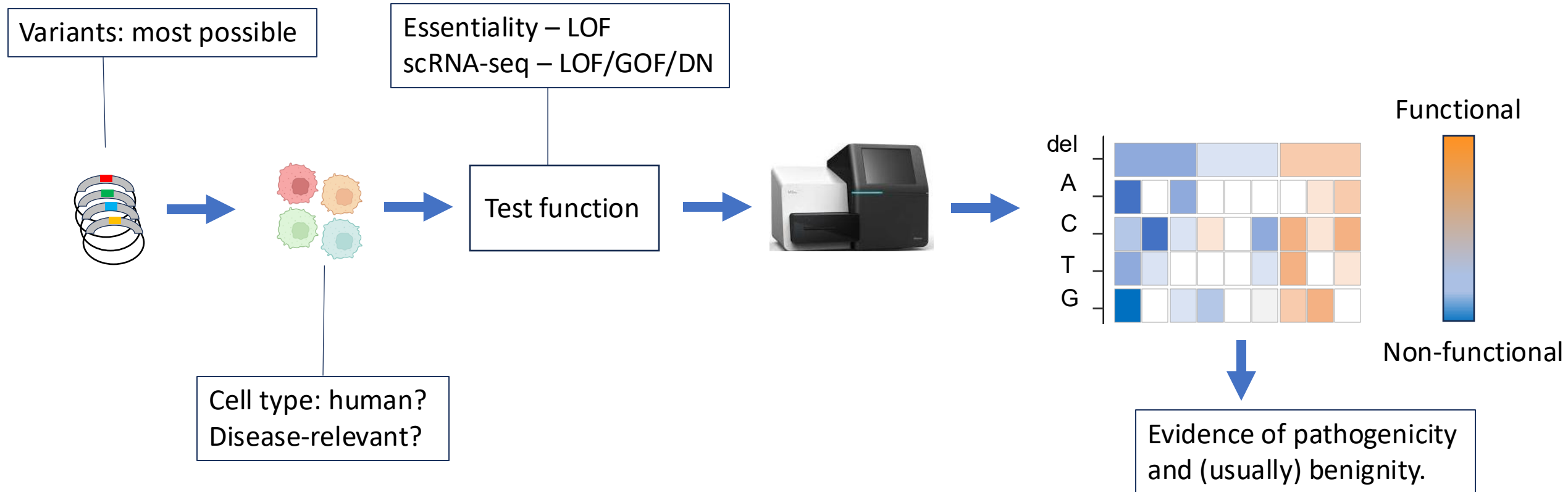
MAVE only measures one aspect of variant function e.g.

- Protein stability
- One aspect of protein function – e.g. binding to a partner

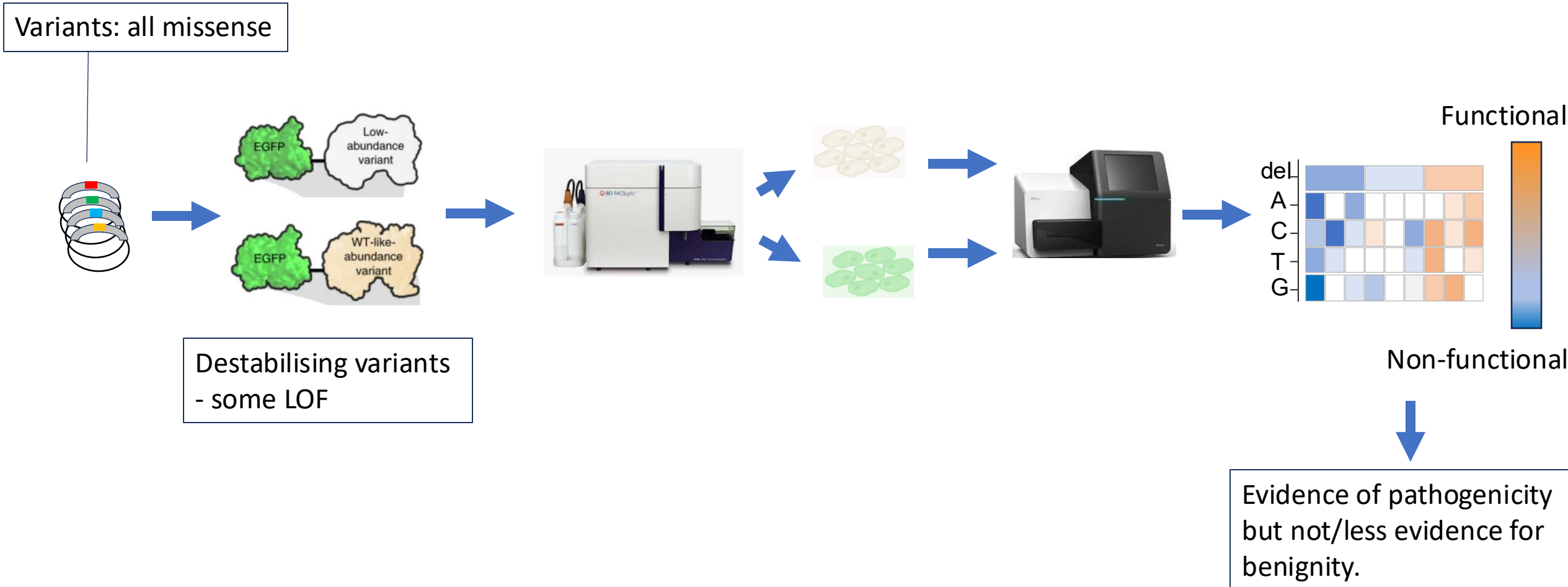


Splice variants are known to be important pathogenic variants, and MAVE does not measure them.

Editing the endogenous locus – e.g. SGE



Measurement of protein stability e.g. vAMP-seq



How do we use MAVE data in variant classification?

Gelman *et al. Genome Medicine* (2019) 11:85
<https://doi.org/10.1186/s13073-019-0698-7>

Genome Medicine

OPINION

Open Access

Recommendations for the collection and use of multiplexed functional data for clinical variant interpretation



Hannah Gelman^{1,2}, Jennifer N. Dines^{1,3,4}, Jonathan Berg⁵, Alice H. Berger^{6,7}, Sarah Brnich⁵, Fuki M. Hisama^{3,7}, Richard G. James^{7,8,9}, Alan F. Rubin^{10,11,12}, Jay Shendure^{1,7,13}, Brian Shirts^{7,14}, Douglas M. Fowler^{1,7,15*}, Lea M. Starita^{1,7*}  On behalf of the Brotman Baty Institute Mutational Scanning Working Group

GUIDELINE

Open Access

Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework



Sarah E. Brnich¹ , Ahmad N. Abou Tayoun², Fergus J. Couch³, Garry R. Cutting⁴, Marc S. Greenblatt⁵, Christopher D. Heinen⁶, Dona M. Kanavy¹, Xi Luo⁷, Shannon M. McNulty¹, Lea M. Starita^{8,9}, Sean V. Tavtigian¹⁰, Matt W. Wright¹¹, Steven M. Harrison¹², Leslie G. Biesecker¹³, Jonathan S. Berg^{1*} and On behalf of the Clinical Genome Resource Sequence Variant Interpretation Working Group

How do we use MAVE data in variant classification?

5. Allocate a weight for the data for use within the ACMG/AMP framework (PS3/BS3)

How does the assay perform on known likely pathogenic and likely benign variants?

- Brnich *et al.* 2020 – Odds of Pathogenicity
- Use the DECIPHER tools to help you

GUIDELINE

Open Access

Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework



Sarah E. Brnich¹ , Ahmad N. Abou Tayoun², Fergus J. Couch³, Garry R. Cutting⁴, Marc S. Greenblatt⁵, Christopher D. Heinen⁶, Dona M. Kanavy¹, Xi Luo⁷, Shannon M. McNulty¹, Lea M. Starita^{8,9}, Sean V. Tavtigian¹⁰, Matt W. Wright¹¹, Steven M. Harrison¹², Leslie G. Biesecker¹³, Jonathan S. Berg^{1*} and On behalf of the Clinical Genome Resource Sequence Variant Interpretation Working Group

How do we use MAVE data in variant classification?

NB difficulties with Brnich *et al.*

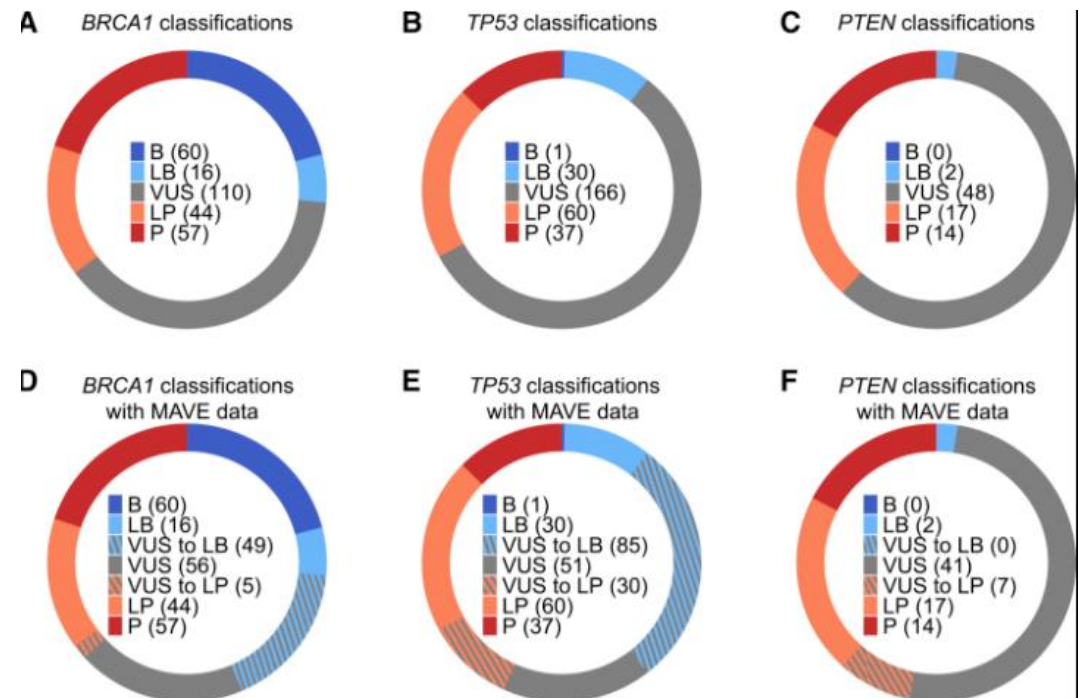
1. Designed for low-throughput functional assays, not for MAVEs
2. Ideally requires a minimum number of pathogenic and benign missense variants
 - Noncoding regions?
3. Can we improve on this with a more MAVE-bespoke approach?
 - Thursday's course!

What to do when there are multiple MAVEs?

1. Get into the nitty gritty of MAVE test, disease mechanism and variant type.
2. VCEP guidance?

Two approaches:

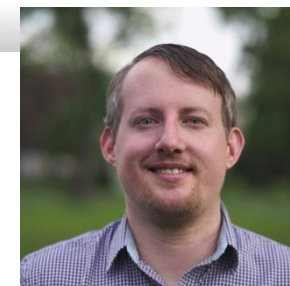
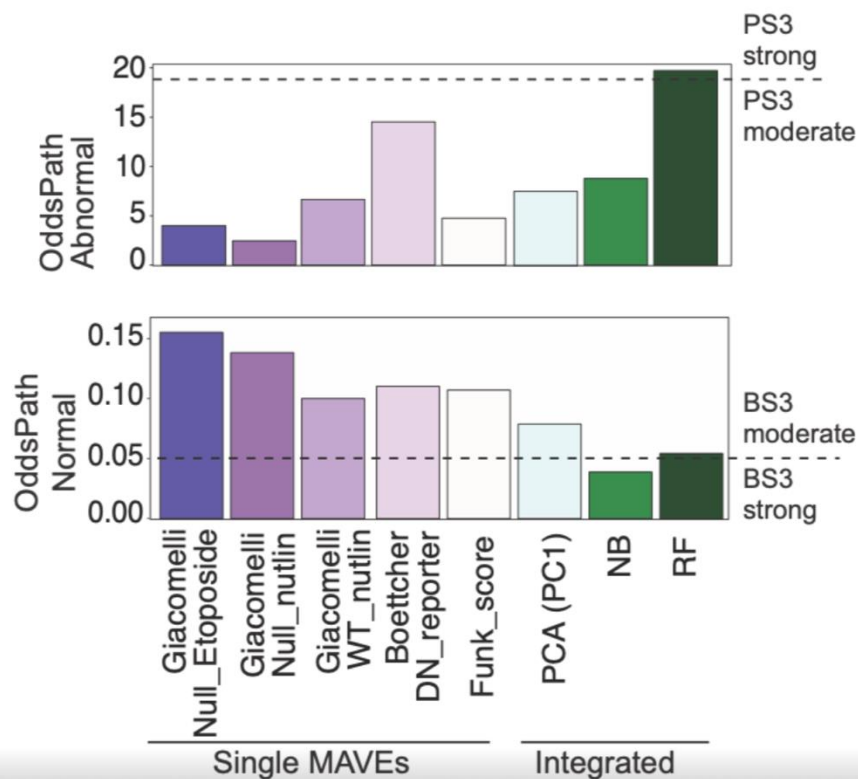
1. Best performing MAVE
2. Combine across MAVES
- Calhoun JD *et al.* ArXiv, 2025



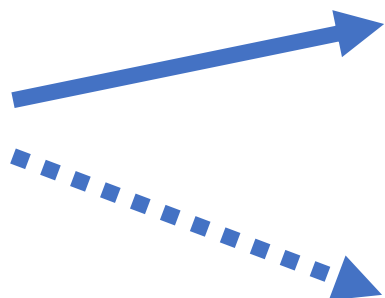
How you combine MAVEs matters

1

TP53	Sens.	Spec.	PPV	NPV	AUC
MAVE1: Etoposide	0.897	0.783	0.627	0.949	0.892
MAVE2: Null_nutlin	0.931	0.629	0.505	0.957	0.825
MAVE3: WT_nutlin	0.931	0.867	0.740	0.969	0.942
MAVE4: DN_reporter	0.914	0.944	0.869	0.964	0.960
MAVE5: Funk	0.931	0.811	0.667	0.967	0.905
PCA	0.948	0.881	0.764	0.977	0.948
Naive Bayes (train)	1.000	0.906	0.773	1.000	0.974
Naive Bayes (test)	0.958	0.865	0.821	0.970	0.944
Random forest (train)	1.000	0.981	0.944	1.000	1.000
Random forest (test)	0.917	0.892	0.846	0.943	0.954



How do I find MAVE data?



CanGene
CanVar

Search...

Contact Us

The Programme CanVIG-UK CanVar-UK ¹³ National Cancer Genetics MDT News and Updates

Functional Studies Scoring

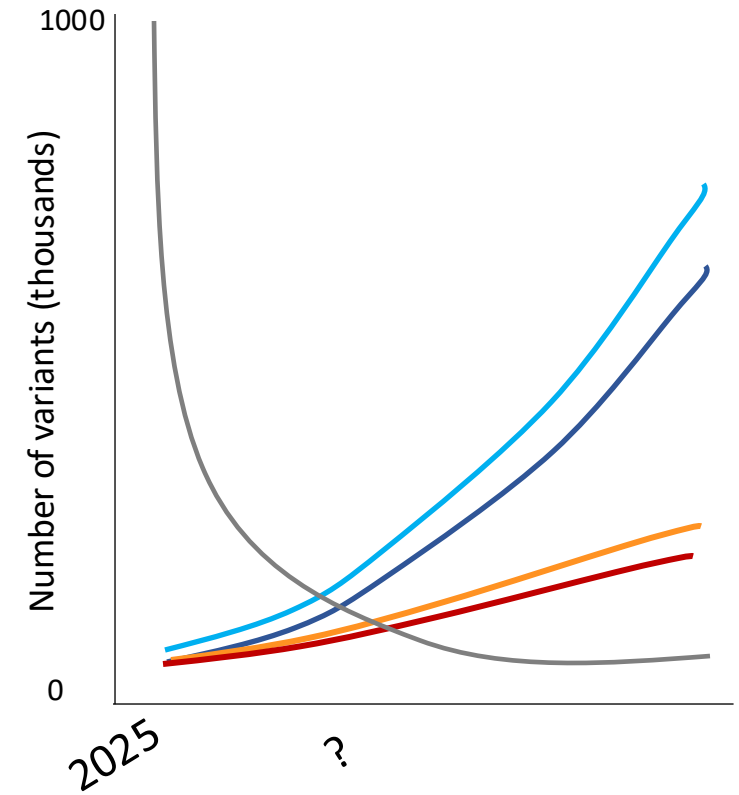
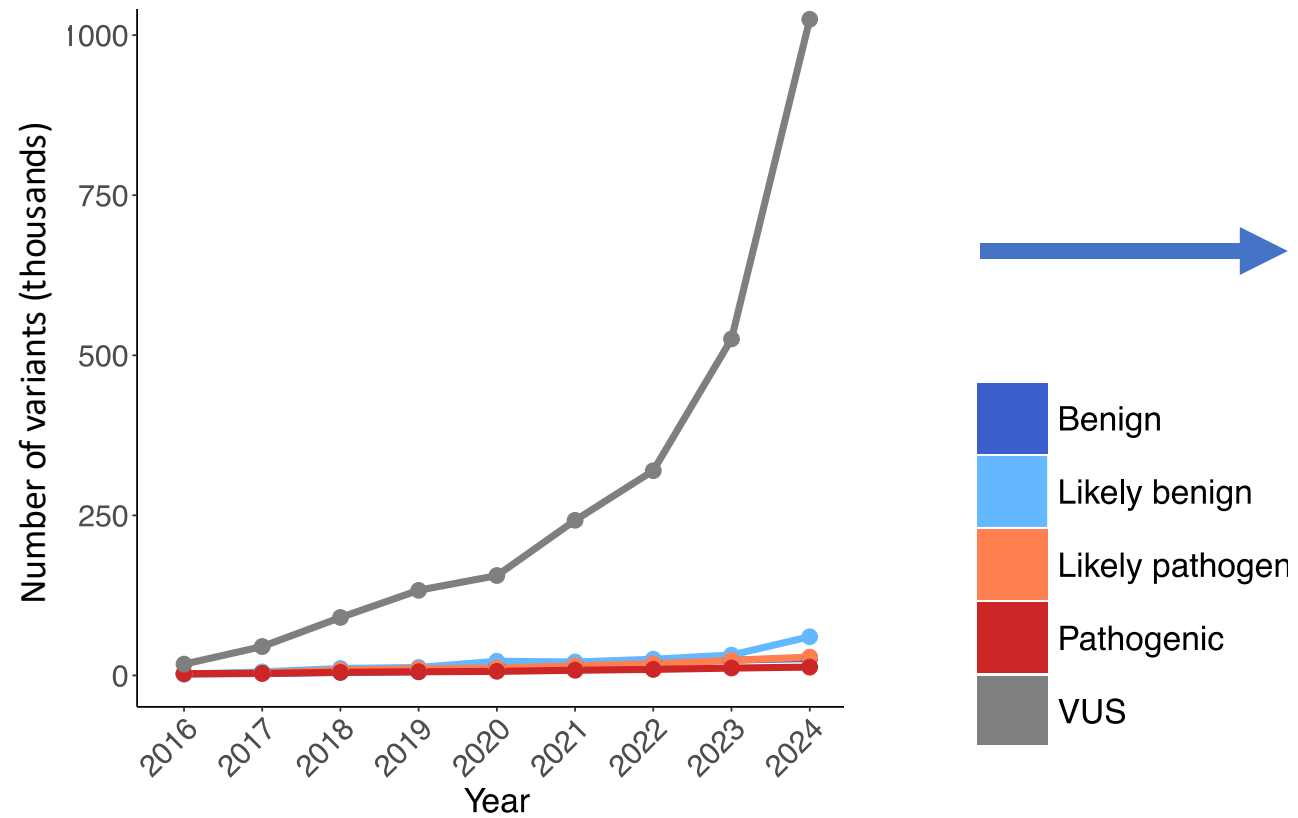
[CanVIG-UK Functional Studies Scoring \(live sync\)](#) [Access Full Scoring Sheet](#)

 **DECIPHER**
GRCh38

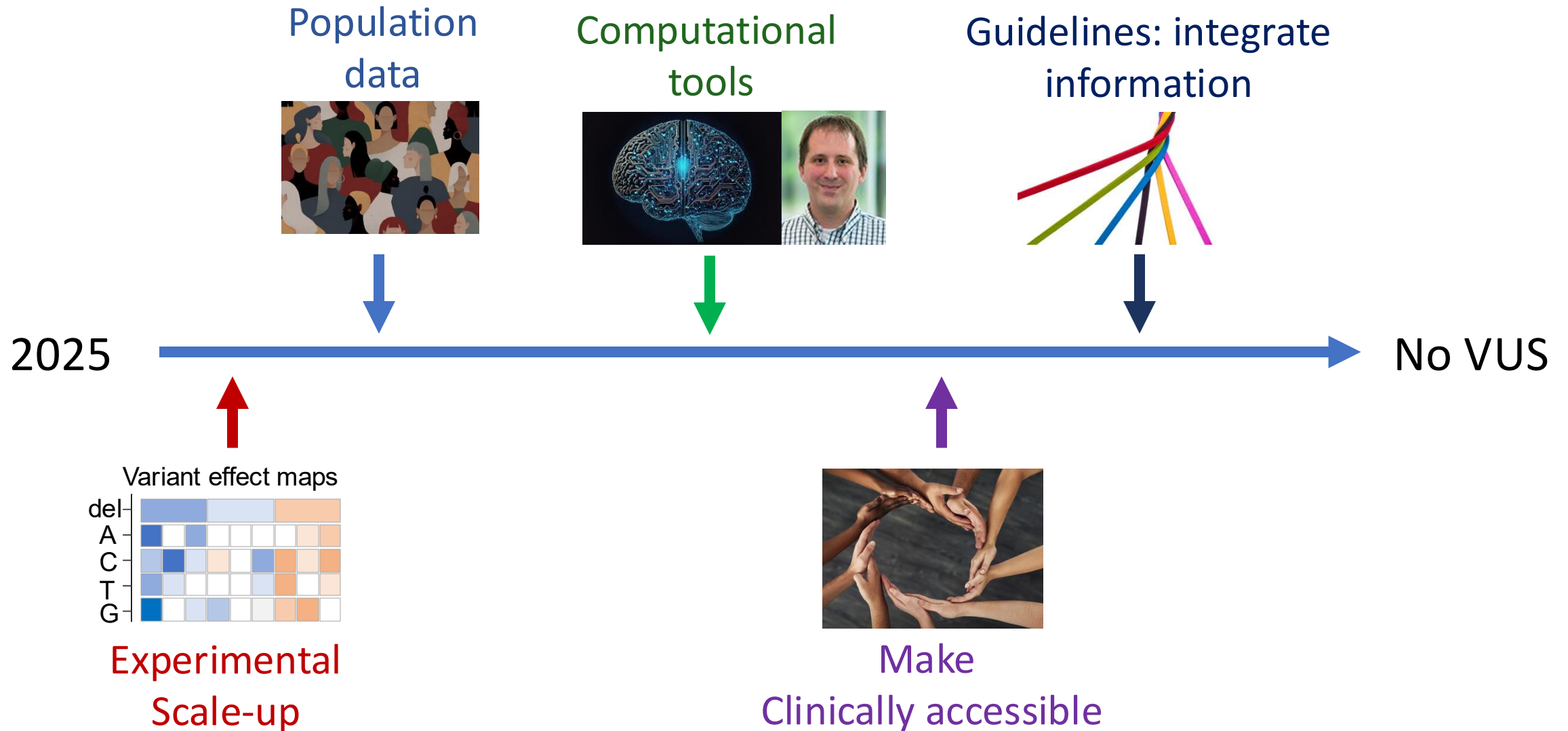
ClinVar
Clinically relevant variation

CTGATGGTATGGGGCCAAGAGATA
AGGTACGGCTGTCACTAGAC
AGGGCTGGGATAAAGTCAAGGC
CATGGTGCATCTGACTCCTGAGGA
CAGGTTGGTATCAAGTTACAAGA
GCAC TGACTCTCTCGCCTATTGG

What's it going to take?



What's it going to take?

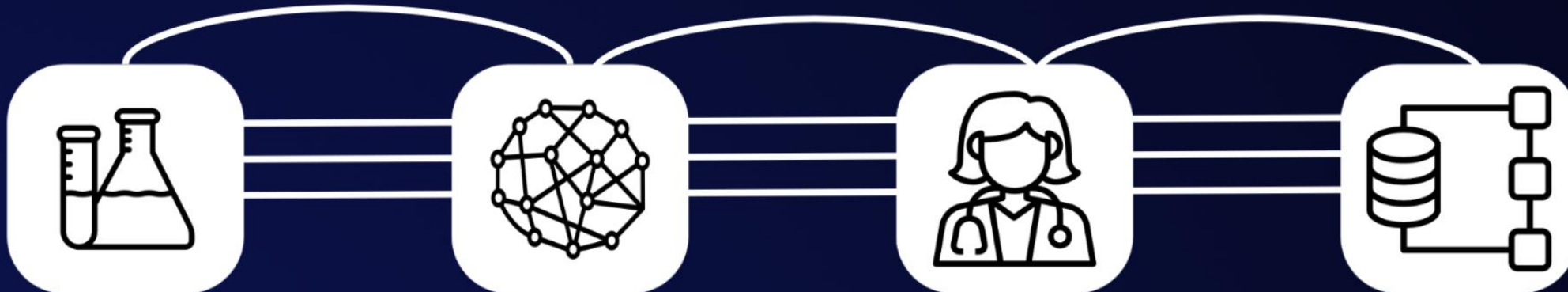


It's going to take a village...

Atlas of Variant Effects Alliance



How we work: 4 interconnected workstreams



Guide

Experimental
Technologies
and Standards
(ETS)



Led by Alex Nguyen Ba &
Andrew Glazer

Analyze-Predict

Analysis,
Modeling and
Prediction
(AMP)



Led by Joseph Marsh

Translate

Clinical
Variant
Interpretation
(CVI)



Led by Lea Starita
& Clare Turnbull

Share

Data
Coordination &
Dissemination
(DCD)



Led by Alan Rubin
& Julia Foreman

MAVE resources <https://www.varianteffect.org/resources:>



[Home](#) [About](#) [People](#) [How we work](#) [Membership](#) [Resources](#) [Seminar](#) [Podcast](#) [Events](#)
[Mutational Scanning Symposium](#) [Donate](#) [Contact Us](#)

MAVE resources

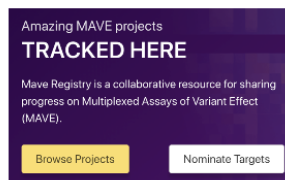
Multiplexed assays of variant effects (MAVEs) allow us to assess (in a single experiment) thousands of variants simultaneously. This technique, which harnesses the power of next-gen sequencing, yields large-scale data sets that can shed light on the functional consequences of protein variants and ultimately on the the landscape of human genetic variation. We are currently working on providing a comprehensive MAVE knowledge base. We hope these preliminary resources will be a helpful starting place!

If you are interested in helping to further develop this resource, please contact program manager Lara Muffley ([muffley \[at\] uw.edu](mailto:muffley@uw.edu))



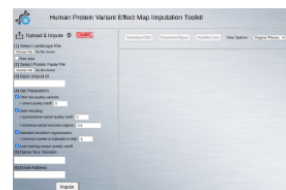
MaveDB

A public repository for variant effect maps



MAVE registry

A collaborative resource for sharing progress on MAVEs



MAVE impute

Impute missing data for variant effect maps

rothlab/**pacybara**

pacbio barcode clustering and barseq pipeline

rothlab/**tileseqMave**

The RothLab TileSeq analysis pipeline for MAVE

MAVE pipelines

[Pacbio barcode clustering](#), [TileSeq analysis pipeline](#) for MAVEs

Thank you:

Matthew Hurles

Sebastian Gerety

Rebecca Yakob

Hong Kee Tan

Malin Andersson

Atlas of Variant Effects Alliance

Alan Rubin

Lea Starita

Clare Turnbull

Joe Marsh

Clinical Variant Interpretation
workstream

Sanger Collaborators:

Dave Adams

Andrew Waters

Victoria Offord

Leopold Parts

Andrew Bassett

Daniel Gitterman

Helen Firth

Cytometry Core Facility

CGaP & Gene Editing

External Collaborators:

Caroline Wright

Emma Baple

Helen Dolling

Next Generation Children Families

Joerg Klepper

GLUT1 Deficiency Foundation

Barth Syndrome Foundation

Funders:



Over to Alan...